CASE STUDY 4 – GROUP 7

MBAN 5540 – Prescriptive Analytics and Optimization

Production Planning at Viktor Lenac Shipyard

Authors

Pamela Ballesteros

Jaiyeoba Tomisin

Instructor

Professor Majid Taghavi

Date

April 2026

PROBLEM BACKGROUND

Viktor Lenac Shipyard was originally founded in 1896 as Lazarus Shipyard before being renamed to honour a Croatian hero of World War II. The company initially built its reputation exclusively on repairs, which formed the backbone of its client relationships and business model. Following a planned $80M modernization program financed by the International Finance Corporation and private banks, Viktor Lenac expanded its product offerings beyond repairs to include conversions, offshore jackets, and newbuilding. However, the modernization effort immediately ran into delays and cost overruns, leaving the yard without the cranes, steel workshops, and equipment needed to fully support newbuilding operations. The resulting high fixed costs eroded the competitive advantage the company had long held in repairs, ultimately pushing Viktor Lenac into bankruptcy. Because repairs took only weeks and required no large upfront cash commitment from clients, they became the company's lifeline through the crisis, reinforcing why repairs remained central to Viktor Lenac's survival even as it attempted to diversify.

CORE PROBLEM

The decision problem is to determine which combination of contracts maximizes total contribution subject to resource constraints, and whether that optimal mix is sufficient to achieve the $30M contribution required to cover fixed costs.

CONTRACT	CONTRIBUTION	STEEL (TONS)	CORROSION PROTECTION (M^2)	DOCK DAYS	LABOR HRS
REPAIR (≤40)	$520,171	170	5,700	9	47,645
CONVERSION (≤4)	$5,014,710	1,100	48,000	83	731,000
OFFSHORE (≤2)	$3,098,844	418	4,200	—	170,609
NEWBUILDING (≤2)	$1,600,150	300	10,000	20	100,426
LIMIT	—	5,500	500,000	700	4,000,000

PROBLEM APPROACH

To address this problem, we an applied Integer Linear Programming Model to identify the optimal contract mix that maximizes Viktor Lenac's total contribution within the shipyard's key resource constraints such as steel, corrosion protection, dock days, and labor. First, the objective of maximizing total contribution is a linear combination of all the decision variables as each contract type contributes a fixed amount of money regardless of the quantity accepted. Secondly, all resource constraints scale proportionally with the number of contracts, satisfying the linearity requirement. Third, the presence of multiple competing constraints and decision variables makes the problem too complex for trial-and-error or intuition-based approaches to produce reliable or optimal solutions.

Furthermore, because contracts can only be accepted in whole numbers, the model is formulated as an Integer Linear Program (ILP) to ensure practical solutions. While Viktor Lenac has historically emphasized repairs, that approach does not necessarily maximize profitability under current resource constraints. An intuition-based selection of contracts would not fully account for the trade-offs created by scarce resources, particularly steel. Integer Linear Programming ensures that the selected contract mix reflects the best overall use of constrained resources and produces a globally optimal solution rather than a locally appealing but suboptimal one.

Decision Variables

- R - number of Repair contracts accepted ($0 \leq R \leq 40$)
- C - number of Conversion contracts accepted ($0 \leq C \leq 4$)
- O - number of Offshore Jacket contracts accepted ($0 \leq O \leq 2$)
- N - number of Newbuilding contracts accepted ($0 \leq N \leq 2$)

Objective Function (Maximize Contribution)

Maximize P = 520,171(R) + 5,014,710 (C) + 3,098,844(O) + 1,600,150(N)

Constraints

Steel: 170 (R) + 1,100 (C) + 418 (O) + 300 (N) ≤ 5,500
Corrosion: 5,700(R) + 48,000(C) + 4,200 (O) + 10,000(N) ≤ 500,000
Dock Days: 9R + 83C + 0O + 20N ≤ 700
Labor Hrs: 47,645R + 731,000C + 170,609O + 100,426N ≤ 4,000,000
Non-negativity: R, C, O, N ≥ 0

RESULTS

The Integer Linear Programming model was solved using both Microsoft Excel Solver and Gurobi/ Python, with these tools producing identical results, which further confirmed and strengthened the validity of the solution we got.

Important Caveat: Excel Solver's Sensitivity Report is generated from the LP relaxation (integer constraints are temporarily removed during sensitivity analysis). As a result, the sensitivity output reflects a fractional LP solution (e.g. C ≈3.69) rather than the integer-optimal plan. The optimal contract quantities are identical between both tools; the shadow prices and reduced costs, however, are drawn from the LP relaxation.

CONTRACT TYPE	OPTIMAL QUANTITY	MAXIMUM AVAILABLE
REPAIR	1	40
CONVERSION	4	4
OFFSHORE JACKET	2	2
NEWBUILDING	0	2

The total optimal contribution achieved, yields a total contribution of $26,776,699, which shows a significant improvement over the current plan of $16,600,000. This represents an increase of approximately $10.2 million which translates to a 61% increment. However, this falls short of the $30,000,000 fixed costs target by $3,223,301, meaning that the optimized product mix alone does not sufficiently achieve financial sustainability.

Resource utilization under the optimal plan is summarized below:

RESOURCE	USED	AVAILABLE	SLACK
STEEL (TONS)	5,406	5,500	94
CORROSION PROTECTION	206,100	500,000	293,900
DOCK DAYS	341	700	359
LABOR HOURS	3,312,863	4,000,000	687,137

RESOURCE UTILIZATION

This is a well detailed analysis of the resource utilization under the optimal plan that show cases important insights about the operations of Viktor Lenac.

Steel is the effective bottleneck, with near-full utilization (5,406 of 5,500 tons used). While not strictly binding due to 94 tons of slack, it is the only resource that meaningfully constrains the solution and carries positive marginal value. Even a slight increase in the steel availability could allow for the shipyard to accept one more *repair* contract, which can generate an extra $520,171 in contribution.

In contrast, the remaining three resources have significant slack. The corrosion protection capacity is utilized only at 41.2% which is 206,100 out of 500,000 m^2 leaving about 293,900 m^2 unused. The dock days are utilized only at approximately 49% which is 341 out of 700 available days leaving about 359 days idle. The labor hours are exhausted at 82.8% which is 3,312,863 out of 4,000,000 hours leaving 687,137 hours unused.

This analysis tells us that the shipyard is not experiencing shortage of workforce, dock days or corrosion protection, the main barrier is the limited steel supply. This finding has direct implications for the management strategy which will be discussed further.

SENSITIVITY ANALYSIS AND WILLINGNESS-TO-PAY

To determine what Viktor Lenac should pay for additional resources, we ran sensitivity analysis on the LP relaxation (standard practice, since Excel Solver drops integer constraints for sensitivity output). Steel is the effective bottleneck, with a shadow price of $4,558.83/ton and an allowable increase of 336 tons which means the yard should pay up to that amount above normal procurement cost per additional ton, as that is the incremental contribution it unlocks. All other shadow prices are zero; additional corrosion, dock, or labour capacity generates no value under the current plan.

INTERPRETATION OF RESULTS

The optimal solution that was produced by our model provides several important insights into the production strategy and financial position of Viktor Lenac.

Why conversions and offshore jackets dominate the optimal plan

The model allocates the maximum available contracts to both conversions (4) and offshore jackets (2) because these are the highest-contribution contract types, generating $5,014,710 and $3,098,844 per contract, respectively. A key driver of this result is contribution efficiency per unit of steel, the most constrained resource. Compared with repairs and newbuilding, conversions and offshore jackets generate more contribution per ton of steel, which is why the model prioritizes them. This shows that the optimal plan is driven not only by total contribution, but also by how efficiently each contract uses scarce resources.

Why only one repair contract

After allocating resources to higher-value contracts, only one repair is selected because steel remains the limiting factor. Additional repairs, while profitable, are less efficient in terms of contribution per unit of steel compared to conversions and offshore jackets.

Why zero new buildings

Newbuilding delivers the lowest contribution per unit of resources consumed. After higher-value contracts are allocated, the remaining 264 tons of steel cannot accommodate even one new building (which requires 300 tons), therefore zero newbuilding are selected in the optimal plan.

The $3.2 million gap- the key finding

Even under the optimal production plan, Viktor Lenac generates $26,776,699, which remains $3,223,301 below the $30M fixed-cost target. This is the most important finding of the analysis: operational optimization alone cannot restore financial sustainability. The result points to a structural imbalance between the company's cost base and its current revenue-generating capacity, meaning that both operational improvements and broader strategic actions are required.

RECOMMENDATIONS

Based on the findings of this analysis, the following recommendations are proposed to Viktor Lenac's management, ranked in order of priority and potential impact:

1. ### *PRIORITIZE SECURING ADDITIONAL STEEL SUPPLY*
   To unlock a second Repair contract (which requires 170 tons), Viktor Lenac needs 76 additional tons beyond current supply (170 − 94 = 76). At the shadow price of $4,558.83/ton, the maximum premium for those 76 tons is $346,471 which is well below the $520,171 contribution it unlocks. Management should negotiate a higher allocation with its Czech Republic supplier or explore alternatives of equivalent quality. Larger increases, up to 336 additional tons, could further enable additional high-value contracts.

2. ### *ACTIVELY PURSUE MORE HIGH-VALUE CONTRACTS*
   The optimal plan is influenced not only by resource limits but also by the availability of contracts. Currently, only 4 Conversions and 2 Offshore Jackets are available, which restricts the model's ability to generate higher returns. Since these contract types deliver the highest contribution relative to the resources they use, management should focus on attracting more of them through targeted business development efforts. Expanding the number of available high-value contracts would allow the model to achieve a higher overall contribution and improve the company's financial performance.

3. ### *ADDRESS FIXED COSTS ALONGSIDE REVENUE GROWTH*
   The analysis shows that even with the optimal production plan, the company still falls $3.2M short of its $30M breakeven target. To address this, management should consider steps to reduce fixed costs. This could include renegotiating existing financial obligations, improve operational efficiency, and delay non-essential investments until revenue improves. Acting on costs alongside efforts to increase revenue will be important for achieving long-term financial stability.

4. ### *RECONSIDER NEWBUILDING AS A SHORT-TERM PRIORITY*
   Although Newbuilding is seen as important for long-term growth, the analysis shows that under current conditions, it delivers the lowest return while using a large share of resources. Given the existing constraints, especially on steel, it is not efficient to allocate resources to this type of contract in the short term. Newbuilding can be reconsidered once the company's financial position improves and resource limitations are less restrictive.

STRATEGIC OUTLOOK – BUILDING LONG TERM SUSTAINABILITY

The findings of this analysis show that Viktor Lenac's financial challenges are not simply due to poor contract choices, but rather a deeper imbalance between its fixed costs and its ability to generate revenue. While optimizing the production plan helps address the immediate shortfall, long-term stability

will depend on building more flexibility and resilience into the shipyard's operations and planning. This case demonstrates that resource constraints, particularly steel, not only limit production but actively shape the firm's optimal strategy. As such, operational decisions and procurement strategy must be aligned, as increasing access to critical resources can directly expand the feasible solution space and improve profitability.

In the medium term, completing Phase 2 of the modernization program will put Viktor Lenac in a stronger position to operate more efficiently, handle more work, and compete for higher-value international contracts. This would allow the shipyard to take on a wider range and greater number of contracts, ultimately increasing its overall earning potential beyond what is currently possible with existing resource limits.

To avoid facing a similar situation in the future, management should make quantitative production planning a regular part of decision-making. Instead of relying on experience or intuition, the Integer Linear Programming model developed in this analysis should be updated and used consistently, considering changes in resources, contract availability, market conditions, and costs. This will help ensure that production decisions are based on data and remain aligned with the shipyard's financial goals.

Finally, Viktor Lenac should aim to build a more balanced and diversified mix of contracts over time, reducing its reliance on any single type of work. Past events, such as the decline in newbuilding demand in the early 1990s, highlight the risks of depending too heavily on one market segment. A more balanced contract mix, supported by ongoing analysis and a stable cost structure, will be key to the shipyard's long-term stability.

CONCLUSION

Viktor Lenac Shipyard's current production plan generates about $16.6M in contribution, which is well below the $30M breakeven target. This analysis used Integer Linear Programming to assess whether a better mix of contracts could close this gap.

The optimal solution selecting 4 Conversions, 2 Offshore Jackets, 1 Repair, and no Newbuildings increases contribution by 61% to approximately $26.8M. While this is a significant improvement, it still leaves a shortfall of $3.2M. This confirms that improving the production mix alone is not enough to fully address the company's financial challenges. Steel was identified as the main limiting factor, restricting how much value the shipyard can generate under current conditions. Management should adopt the optimized production plan immediately to capture the available improvement in contribution. At the same time, longer-term actions like securing additional steel supply, reducing fixed costs, expanding the high-value contract pipeline, and completing Phase 2 modernization will be necessary to close the remaining gap and achieve full financial sustainability.

As a further extension, steel procurement could be modeled explicitly as a decision variable in future planning periods. Because the case states that steel must be ordered and shipped in advance, making annual steel availability a planning decision rather than a fixed input, the model could be extended into a Mixed Integer Linear Program (MILP). This would allow the firm to jointly optimize contract selection and steel procurement, using the shadow price of steel as a benchmark for whether additional steel is economically justified.

REFERENCES

Sinković, G., & Currie, D. M. (2017). *Production Planning at Viktor Lenac Shipyard* (Case No. 9B04D019). Ivey Publishing, Richard Ivey School of Business Foundation, Western University.